

12 (**long**) An even function $f(t)$ is periodic with period $T = 2$ and $f(t) = \cosh(t - 1)$ for $0 \leq t \leq 1$.

(a) Sketch $f(t)$ in the range $-4 \leq t \leq 4$. [5]

(b) Show that

$$f(t) = \sinh(1) \left[1 + 2 \sum_{n=1}^{\infty} \frac{\cos(n\pi t)}{1 + n^2\pi^2} \right]$$

is a Fourier series representation of $f(t)$. [15]

(c) Deduce that

$$\sum_{n=1}^{\infty} \frac{1}{1 + n^2\pi^2} = \frac{1}{e^2 - 1}$$

using the solution in (b). [10]

END OF PAPER